

CalSoft, Module Reference

What the calibration module contains, and how a certificate is made

CalSoft is the part of Cirqen that turns manual readings into a numbered calibration certificate. This document covers what it is built from, and follows one certificate from the moment a technician opens a session to the moment a PDF is handed over.

The mathematics behind the numbers is in *The Mathematics of Calibration*; this document is about the machinery around it.

1. The measurement hierarchy

Five levels, each narrowing the measurement. Everything else in the module hangs off this.

Procedure → Parameter → Sub-parameter → Set value → Reading

Level	What it is	Example
Procedure	The method. Owns the environmental conditions the calibration should be performed under.	"NIBP calibration"
Parameter	What is measured. Carries the unit, the tolerance, the reference standard's uncertainty, the coverage factor and how many readings to take.	Systolic pressure, mmHg, ± 3
Sub-parameter	An optional split within a parameter, where one measurement has two channels.	Systolic / diastolic
Set value	A test point: the nominal value applied by the reference.	100, 150, 200, 250, 300 mmHg
Reading	Up to ten raw values at one test point, plus every computed result.	201, 202, 201, 203, 202

Resolution is not on this list, and that is deliberate. It belongs to the device on the day, not to the procedure, the same procedure may be used on a monitor reading whole mmHg and another reading tenths. It is stored per session in `SessionParameterResolution`, and a session with no resolution recorded for a parameter cannot compute that parameter's uncertainty.

2. The session

A **session** is one event: one device, one procedure, one moment, one technician, one set of environmental conditions. It carries the device identity (serial, model, manufacturer, description), the department and workshop, the actual temperature, humidity and pressure at the time, the overall verdict, and, once HQ has issued one, the certificate number.

2.1 Session states

State	Meaning
pending_review	Readings captured; awaiting a reviewer.
approved_pending_certificate	Approved. A certificate number has been requested from HQ and not yet issued.
approved	Approved and numbered. A certificate can be issued.
rejected	Declined, with a reason and comments recorded.

approved_pending_certificate is a real, ordinary state rather than an error. Certificate numbers are allocated by the HQ server only, so there is always a moment between approval and issue.

3. How a certificate is made

3.1 Capture

The technician selects equipment and a procedure, records the environmental conditions, and enters readings for every test point. Environmental conditions are compared against the procedure's requirements and any deviation beyond 2 °C, 10 %RH or 1 kPa is surfaced as a warning, recorded, not blocking.

As readings are typed, the live preview calls the same uncertainty calculation the saved record will use, so the figures on screen and the figures stored are computed identically.

3.2 Reduction

On submit, each test point is reduced to a complete uncertainty budget:

$$\bar{X}, S, U_A, U_{\text{res}}, U_{\text{ref}}, U_C, U, k, v_{\text{eff}}$$

The coverage factor is **derived** from the effective degrees of freedom rather than assumed, with the parameter's configured factor acting as a floor. The deviation from nominal is computed and compared with the tolerance.

The session's verdict is **strict conformity**: a single test point outside tolerance fails the session. A point left blank also fails it, because an unmeasured point is not a passed one.

3.3 Review and approval

A reviewer with the approve_session permission approves or rejects. On approval the session moves to approved_pending_certificate, a PendingCertificate row is queued, and the linked schedule is advanced.

No certificate number is allocated locally. HQ is the only allocator. The field application never mints one, because two sites allocating from their own local sequences produce the same number and collide on sync.

3.4 Issue

HQ allocates the next number, writes it to the session and flips the status to approved. The client syncs it down, and the PendingCertificate row is marked complete. A recovery loop re-requests any number that has not arrived.

3.5 Rendering

The PDF is assembled from eight composable parts, in this order:

Section	What it carries
Title and number	Certificate or reference number, issue date
Three-section table	Device, customer and calibration details
Calibrator & standard	Which reference standards were used, and their certificates
Linearity analysis	Slope, intercept, largest deviation from the fitted line, and a residual chart
Drift analysis	Historical drift per parameter, graded against tolerance
Results & statistics	Per point: set value, mean, standard deviation, error, tolerance, TUR, verdict
Failure analysis	Only when points failed: per-parameter failure rates and recommendations
Uncertainty budget	Per point: n, Type A, Type B, Reference, Combined, Expanded, k
Traceability	The unbroken chain to SI
Notes	Decision rule, coverage factor, measurement-capability warnings
Signatures	Who performed and who approved
Footer	Identity and QR reference

A watermark is drawn over the content on every page, and a QR code carries a summary for reference.

3.6 What the verdict means on the page

Two independent things are reported, and they are not the same:

- **Conformity**, did the device pass? Strict: one failed point fails it.
- **Failure rate**, how widespread is the failure? Above 40 % indicates a device fault rather than isolated drift, and changes the wording of the notes.

The rate is a triage signal for the workshop. It never decides conformity.

4. The parts of the module

4.1 Data model

Model	Holds
CalibrationProcedure	The method, environmental requirements, equipment mappings
CalibrationParameter	Unit, tolerance, reference uncertainty, coverage factor, reading count
SubParameter	An optional channel split, with its own tolerance
SetValue	A test point
CalibrationSession	One calibration event and its verdict
SessionParameterResolution	The resolution in force for a parameter in that session
CalibrationReading	Ten raw slots plus nine computed results

Model	Holds
HistoricalCalibration	A flattened copy of every computed reading, for drift
PendingCertificate	A queued request for a number from HQ
CalibrationAuditLog	Who did what, in readable form
Standard, StandardParameter	Reference standards and their accredited uncertainties

4.2 The calculation engine

One module, pure arithmetic, no database, no framework, so it can be tested directly. It provides the statistics, the three uncertainty components, the combination and expansion, the effective degrees of freedom, the test uncertainty ratio, the guarded conformity decision, linearity by least squares, drift grading against tolerance, and robust outlier detection.

Every formula in it is pinned to a hand-computed test vector.

4.3 Views

Session capture and submission, review and approval, certificate listing and download, procedure and parameter management, standards import from Excel, a dashboard, and a JSON API used by the live preview.

4.4 The PDF generator

Composed from eight mixins, styles, results, QR, header, sections, analysis, signatures, drift, so each concern is separately readable and separately testable.

5. What a certificate guarantees, and what it does not

Worth stating plainly, because a certificate is a document people rely on.

It does guarantee

- the arithmetic is correct and reproducible from the printed figures;
- the verdict is strict conformity, stated with the decision rule used;
- the coverage factor shown is the one actually applied;
- the number was allocated by one authority and was not invented locally.

It does not yet guarantee

- **tamper-evidence.** The QR code carries a summary for reference. It is not a cryptographic signature, and scanning it does not by itself prove the document is genuine. Confirm a certificate number against the issuing record.
- **a signed approver identity.** The signature block records who approved and when, as an image, not as a cryptographic signature.

These are the outstanding items in the platform plan, and the certificate no longer claims otherwise: the note and footer state what the QR code is for rather than overstating it.

6. Known limits

Recorded here rather than left to be discovered.

Limit	Consequence
Sensitivity coefficients are not modelled	Every component is assumed to act one-for-one on the measurand, so an influence acting indirectly, a temperature coefficient, lead resistance, hydrostatic head, cannot enter the budget.
Type B components are fixed at two	Resolution and reference standard only.
The stored verdict is two-valued	The certificate reports PASS, FAIL and INDETERMINATE; the database stores pass or fail.
Minimum readings are not enforced from below	A parameter declaring five readings will certify on two. Two readings give one degree of freedom and a coverage factor of 12.706, which is correct and not a certifiable measurement.
The deviation sign convention	Confirm the column heading before interpreting the sign; see <i>The Mathematics of Calibration</i> section 9.1.

Cirqen Calibration Software. The mathematics behind these figures is derived in full in the companion volume, The Mathematics of Calibration.

